Nets and surface area

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Use decompositions and nets to count every part of a surface.

R1. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

R2. Find $10 - 2.286$.

R3. Solve $x + 9 = 23$.

R4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

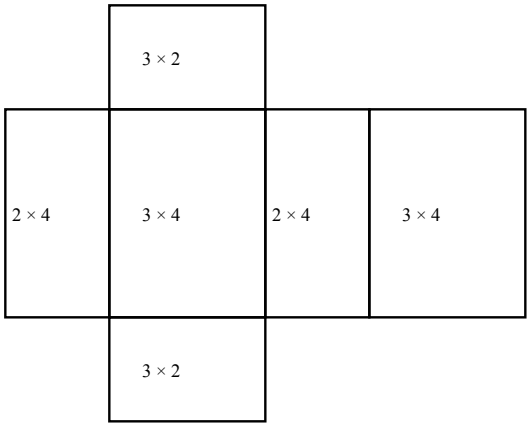
P1. An 8 cm by 6 cm rectangle loses a 3 cm by 2 cm corner from the upper right. Find the L-shaped area two ways.

P2. A closed box is 2 cm by 3 cm by 4 cm. Label all six rectangles on a net. Find surface area and volume.

P3. A triangular prism has right-triangle ends with sides 6, 8, and 10 cm. Its length is 5 cm. List the five faces and find surface area.

P4. A square pyramid has base side 6 cm. Each triangular face has perpendicular height 5 cm. Find its surface area.

Lesson model



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Practice and compare methods

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P5. A closed 2 cm by 4 cm by 5 cm box has a claimed surface area of 68 square cm. Its 2 by 4 bottom was left out. Repair the answer.

F. A square pyramid has base side 4 cm and face height 3 cm. Complete: SA = 16 + 4 × ($\frac{1}{2}$ × 4 × __) = __.

D. Ava finds a cube's surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

M1. A ribbon is $4\frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

M2. Find 312 ÷ 24.

M3. Use y = 2x for x = 1, 3, 5. Give the ordered pairs.

X1. A closed box measures 2 cm by 4 cm by 5 cm. Find surface area.

X2. A cube has side 4 cm. Find surface area and volume; label units.

Worked example - compare with the slide model

Find the surface area of a 2 cm by 2 cm by 3 cm closed box.

Two square ends each have area 4.

Four side rectangles each have area 6.

Total area = 2 × 4 + 4 × 6 = 32 square cm.

Result: 32 square cm

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Challenge, explain and check

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C1. A closed rectangular box is 4 cm by 3 cm by 2 cm. Find its surface area. A second version has no 4-by 3 top and needs an extra 2-by 3 label made from the same material. How much material does the second version need, and how much is saved?

C2. A triangular prism has right-triangle ends with sides 3, 4, 5 cm and length 10 cm. Find surface area and volume. A builder claims that doubling the length doubles both results. Test the claim by using length 20 cm.

Finish the lesson

S. To find surface area, I list __ and add __.

E. A triangular prism has right-triangle ends with sides 3, 4, and 5 cm. It is 8 cm long. Find the area of all five faces together.